

```
import numpy as np
import pandas as pd
```

```
df = pd.read_csv('iris-data.xls')
```

df

	sepal length	sepal width	petal length	petal width	class
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

Next steps:

[Generate code with df](#)
[New interactive sheet](#)

```
X = df.drop(columns='class', axis = 1)
```

```
from sklearn.cluster import KMeans
km= KMeans (n_clusters = 3 , random_state = 1)
km.fit(X)
```

KMeans

KMeans(n_clusters=3, random_state=1)

km.labels_

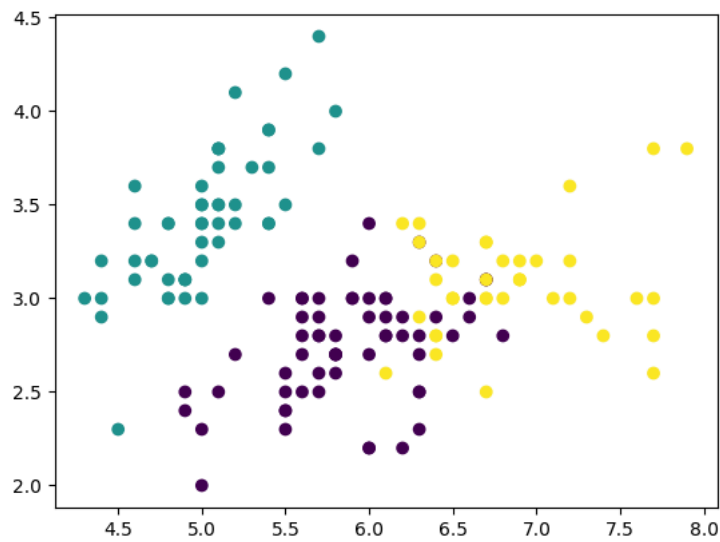
```
array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 2, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 2, 2, 2, 2, 0, 2,
       2, 2, 2, 2, 2, 2, 0, 2, 2, 2, 0, 2, 2, 0, 0, 2, 2, 2, 2,
       2, 0, 2, 2, 2, 2, 0, 2, 2, 2, 0, 2, 2, 2, 0], dtype=int32)
```

km.cluster_centers_

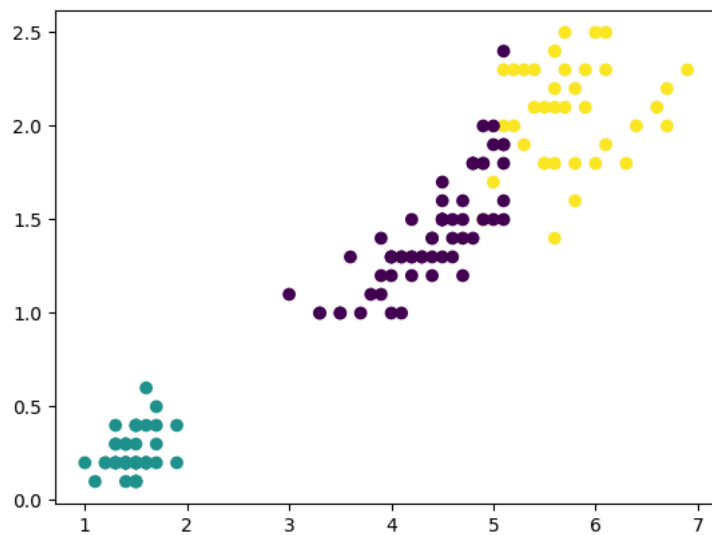
```
array([[5.88360656, 2.74098361, 4.38852459, 1.43442623],
       [5.006      , 3.418      , 1.464      , 0.244      ],
       [6.85384615, 3.07692308, 5.71538462, 2.05384615]])
```

```
import matplotlib.pyplot as plt
```

```
plt.scatter(X['sepal length'], X['sepal width'], c=km.labels_)
plt.show()
```



```
plt.scatter(X['petal length'], X['petal width'], c = km.labels_)
plt.show()
```

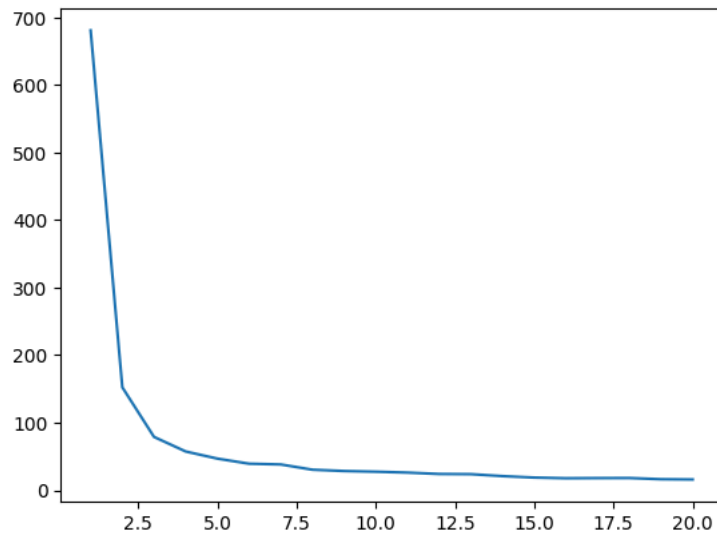


```
inertia_scores = []
for i in range(1,21):
    k = KMeans(n_clusters=i)
    k.fit(X)
    inertia_scores.append(k.inertia_)
```

```
inertia_scores
```

```
[680.82439999999996,
152.36870647733915,
78.94506582597728,
57.34540931571815,
46.80170193050192,
39.266821514027406,
38.15272268170426,
30.367198747307462,
28.41163344988346,
27.40629313166195,
26.09240418894831,
24.037537329577138,
23.76414354066986,
20.857641708816903,
18.72724025974027,
17.72071844821845,
17.947111832611842,
18.040423160173173,
16.28087912087912,
15.905703074703082]
```

```
x_labels= range(1,21)
plt.plot(x_labels, inertia_scores)
plt.show()
```



```
from sklearn.cluster import AgglomerativeClustering
```

```
cluster = AgglomerativeClustering(n_clusters = 3).fit(X)
```

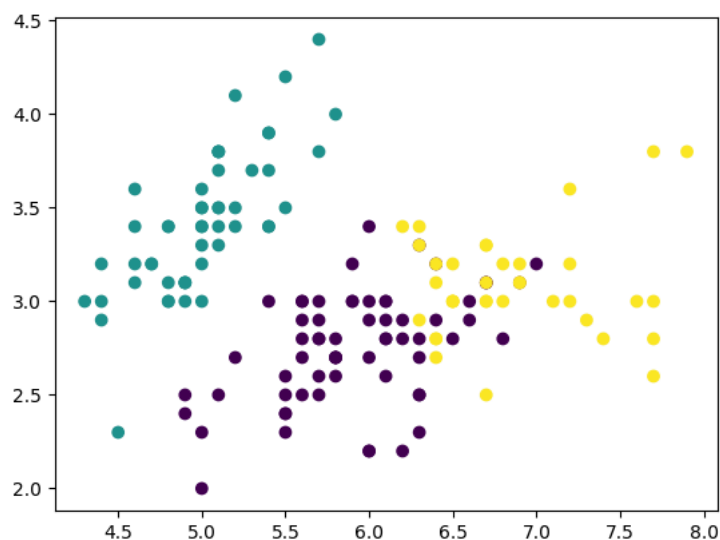
cluster

- AgglomerativeClustering

```
cluster.labels_
```

```
array([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,  
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0,  
       1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 2, 2, 2, 0, 2, 2,  
       2, 2, 2, 0, 0, 2, 2, 2, 2, 2, 0, 2, 0, 2, 0, 0, 2, 2, 2, 2,  
       2, 0, 0, 2, 2, 2, 0, 2, 2, 2, 2, 0, 2, 2, 2, 0])
```

```
plt.scatter(X['sepal length'],X['sepal width'],c= cluster.labels_)
plt.show()
```



```
from sklearn.metrics import confusion_matrix
```

```
confusion_matrix(km.labels_, cluster.labels_)
```

```
array([[61,  0,  0],
       [ 0, 50,  0],
       [ 3,  0, 36]])
```

```
df=pd.DataFrame({'km':km.labels_ , 'ac': cluster.labels_})
```

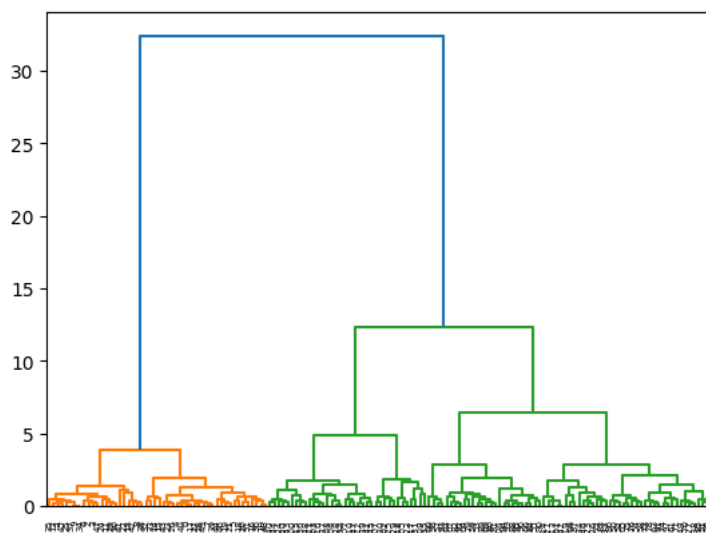
df

	km	ac
0	1	1
1	1	1
2	1	1
3	1	1
4	1	1
...	...	...
145	2	2
146	0	0
147	2	2
148	2	2
149	0	0

150 rows × 2 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
import scipy.cluster.hierarchy as sch
dendrogram = sch.dendrogram(sch.linkage(X, method = 'ward'))
```



```
plt.subplot(2,2,1)
plt.scatter(X['sepal length'], X['sepal width'], c= km.labels_)
plt.subplot(2,2,2)
plt.scatter(X['sepal length'], X['sepal width'], c = cluster.labels_)
plt.subplot(2,2,3)
plt.scatter(X['petal length'], X['petal width'], c = km.labels_)
plt.subplot(2,2,4)
plt.scatter(X['petal length'], X['petal width'], c = cluster.labels_)
```

```
<matplotlib.collections.PathCollection at 0x7e5e145faae0>
```

